Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

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1 Introduction

2 Institutional Background

2.1 Ohio Municipal Income Tax

2.2 Pennsylvania Earned Income Tax

3 Data

3.1 LODES Origin-Destination Data

3.2 Municipal Tax Rate Panels

3.3 Municipal Financial Reports

3.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	24,528 (1,383 municipalities)				
OH municipality × years	7,718 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

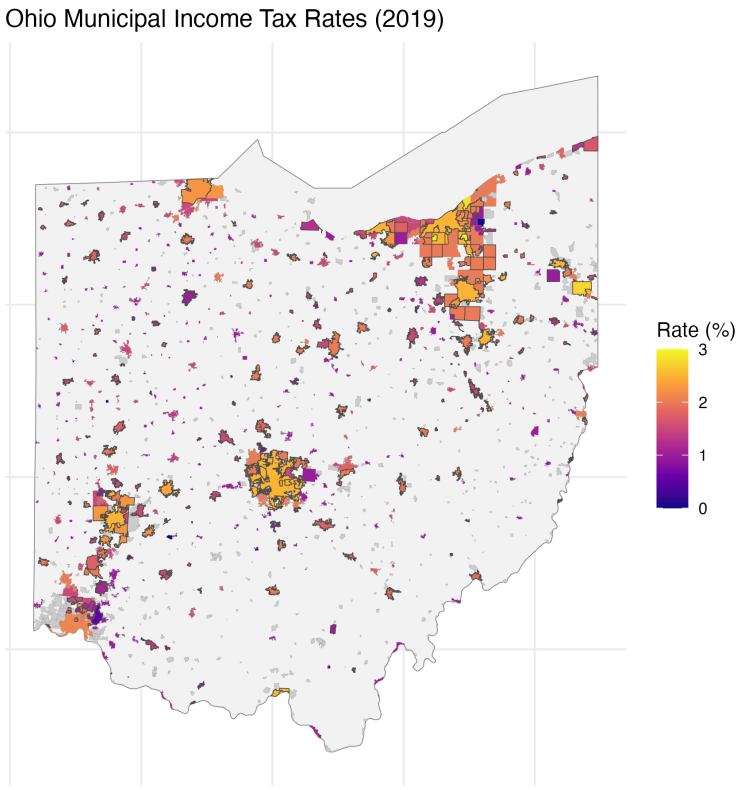


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

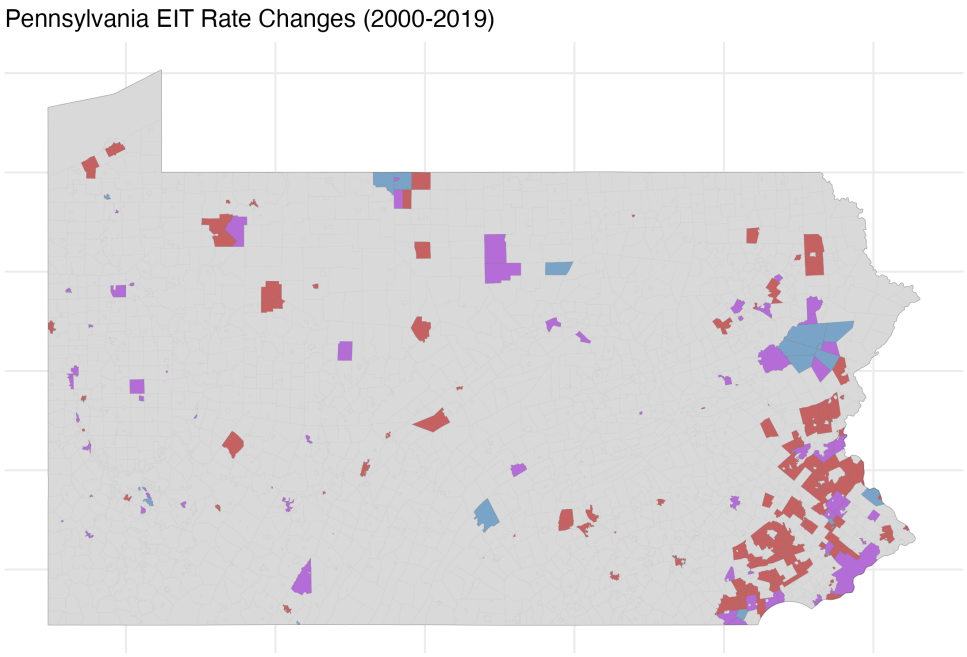


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

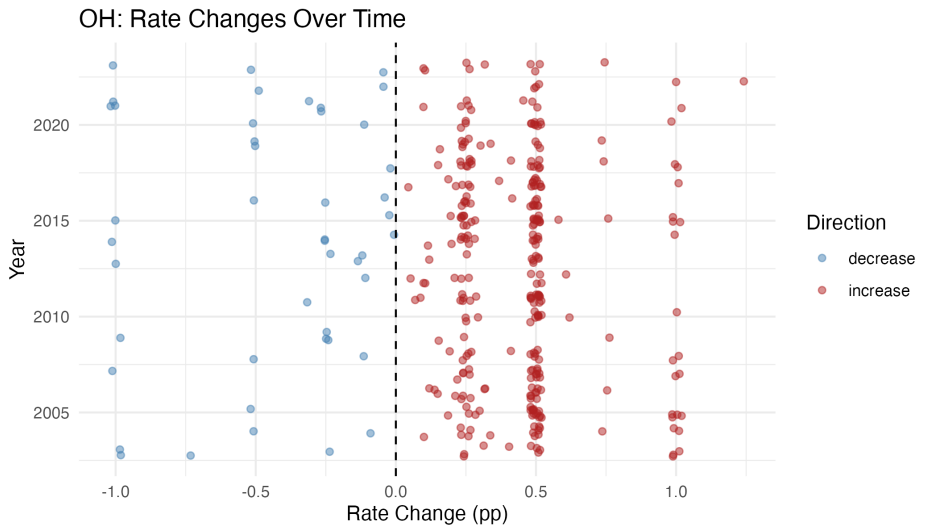


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

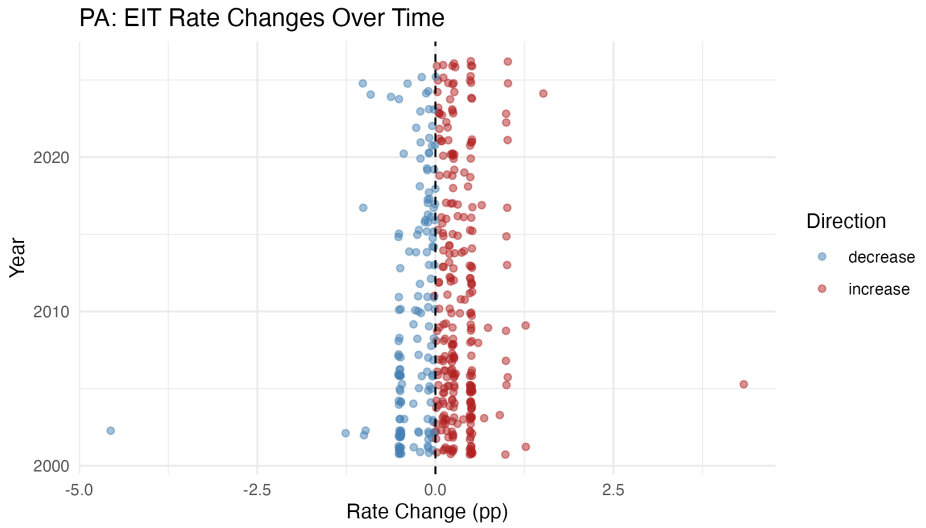


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

4 Empirical Strategy

4.1 Baseline Specification

$$Y_{it} = \alpha_i + \gamma_t + \beta \cdot \text{rate}_{it} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{1}$$

where:

Y_{it} = share_live_in for municipality i in year t

α_i, γ_t = municipality and year fixed effects

rate_{it} = municipal income tax rate (EIT in PA, municipal rate in OH)

$\mathbf{X}_{it}$ = vector of fiscal controls: log police expenditure, log streets expenditure, log total debt, log state intergovernmental aid

Standard errors clustered at the municipality level

4.2 Dose-Response Identification

$$Y_{it} = \alpha_i + \gamma_t + \delta \cdot \text{post}_{it} + \beta \cdot (\text{rate}_{it} \times \text{post}_{it}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

where:

$\text{post}_{it} = 1[t \geq g_i]$, with g_i the year of municipality i's first rate increase

δ captures common post-treatment shifts among rate-changing municipalities

β captures the tax-specific channel: conditional on raising taxes, does rate magnitude predict additional changes in the locally-resident share?

$\mathbf{X}_{it}$ = fiscal controls as in Equation **??**

Sample restricted to municipalities that raised rates during the panel

4.3 Sun-Abraham Event Study

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot 1[t - g_i = \ell] + \varepsilon_{it} \tag{3}$$

where:

g_i = cohort (year of first rate increase for municipality i; $g_i = \infty$ for never-treated)

$\ell = t - g_i$ = relative time to treatment

μ_ℓ = cohort-weighted average treatment effect at relative time ℓ, estimated via the interaction-weighted estimator of Sun and Abraham (2021)

$\ell = -1$ is the reference period (normalization)

Pre-treatment coefficients μ_ℓ for $\ell < 0$ test the parallel trends assumption

4.4 Stacked Difference-in-Differences

$$Y_{ict} = \alpha_{ic} + \gamma_{ct} + \sum_{\ell \neq -1} \beta_\ell \cdot (\text{treated}_i \times 1[\ell_{ict} = \ell]) + \varepsilon_{ict} \tag{4}$$

where:

c indexes cohort (one per treatment event)

α_{ic} = cohort-specific municipality fixed effects

γ_{ct} = cohort-specific year fixed effects

$\text{treated}_i = 1[i \text{ is the treated municipality in cohort } c]$

$\ell_{ict} = t - g_c$ = relative time within cohort c

Control municipalities: neighbors within 80km of the treated municipality that did not change their rate within ± 2 years of the event

OH specification: control group further restricted to municipalities with no income tax (unincorporated townships)

Event window: $\ell \in \{-5, \ldots, +5\}$

Standard errors clustered at the municipality level

5 Results

5.1 Main Results

Table 2: Main Results: Effect of Local Income Tax Rate on share_live_in

	(1) PA TWFE	(2) PA Dose	(3) PA Post + Dose	(4) OH Stacked t+5
Coefficients				
resident_eit	-0.0101^{*}			
	(0.0050)			
rate × post		-0.0084^{**}	$-0.0099^{\cdot}$	
		(0.0032)	(0.0052)	
post			0.0022	
			(0.0064)	
treated × t+5				-0.0109^{***}
				(0.0026)
Diagnostics				
N	24,522	2,741	2,741	48,712
R^2	0.858	0.912	0.912	0.922
Within R^2	0.0004	0.010	0.011	0.002
Dep. var. mean	0.149	—	—	—
Fixed Effects	City, Year	City, Year	City, Year	Cohort×City, Cohort×Year
Clusters	Municipality	Municipality	Municipality	Municipality
Sample	All PA	Treated PA	Treated PA	OH border (no-tax controls)

Notes: Dependent variable is share_live_in (fraction of workers in municipality who also reside there). Column (1): two-way fixed effects on all PA municipalities. Columns (2)–(3): dose-response among PA municipalities that raised EIT rates, testing whether rate magnitude predicts sorting beyond common post-treatment shifts. Column (4): stacked difference-in-differences for Ohio, comparing treated cities to neighboring municipalities with no income tax within 80km. Standard errors clustered at the municipality level in parentheses. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

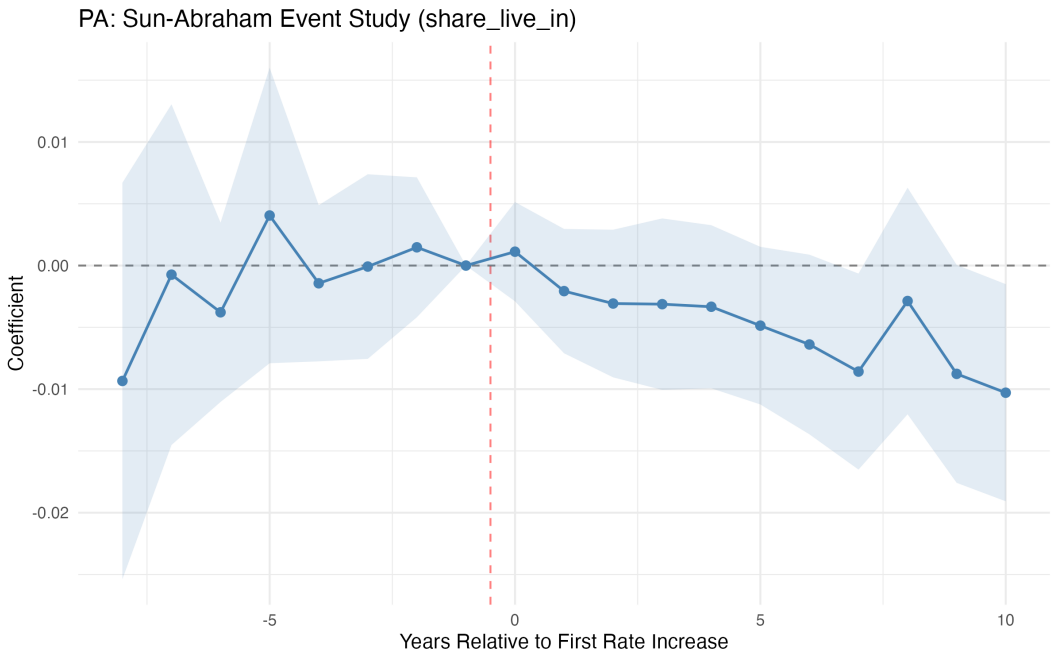


Figure 5: Pennsylvania: Sun-Abraham Event Study (share_live_in). Cohort defined by first year of EIT rate increase. Never-treated municipalities as control group. Shaded region = 95% confidence interval.

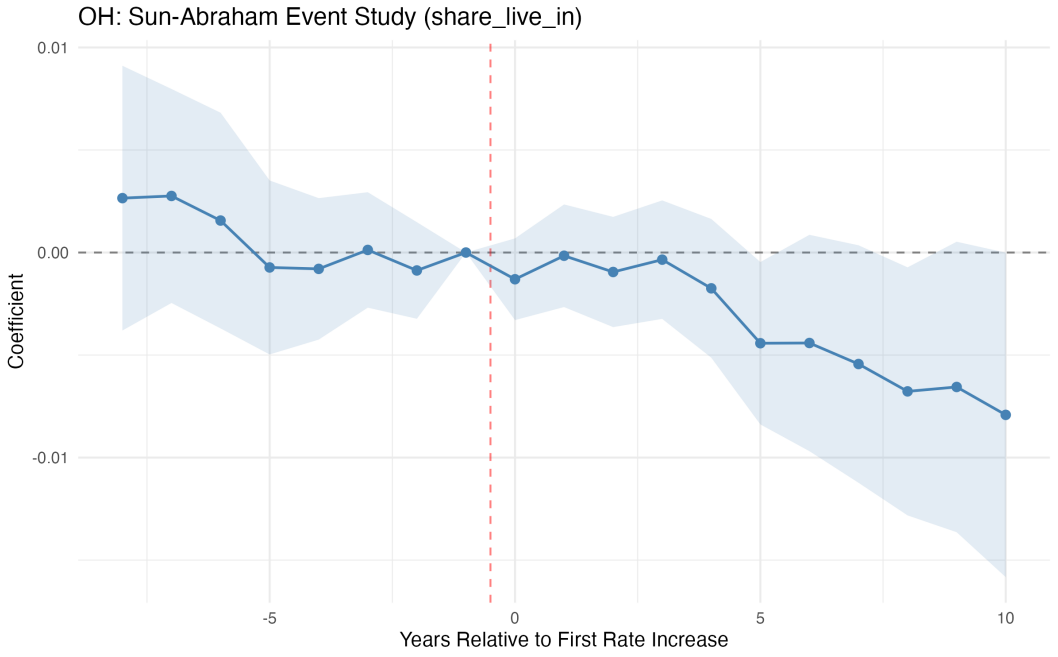


Figure 6: Ohio: Sun-Abraham Event Study (share_live_in).

5.2 Companion Outcomes: Decomposition

Table 3: Companion Outcomes: Decomposing the share_live_in Effect

	(1)	(2)	(3)	(4)
	share_live_in	log(Resident)	log(Nonresident)	log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	−0.0099·	−0.1530*	−0.1012·	−0.1172*
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	0.1226·	0.1113·	0.1173*
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
Diagnostics				
N	2,741	2,741	2,741	2,741
R^2	0.912	0.979	0.980	0.982
Within R^2	0.011	0.015	0.008	0.011
Fixed Effects		Municipality, Year		
Clusters		Municipality		
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification estimated among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts of resident workers (live and work in municipality), nonresident commuters (work in municipality, live elsewhere), and total jobs. Equality test: joint model testing whether the resident coefficient differs from the nonresident coefficient. $\cdot p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

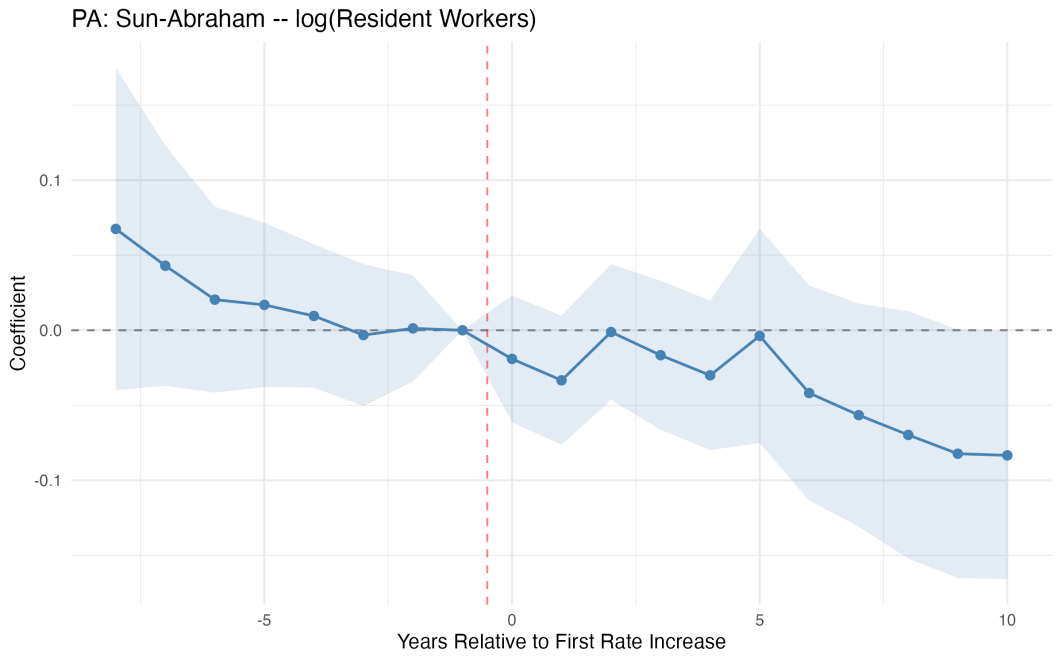


Figure 7: Pennsylvania: Sun-Abraham Event Study — log(Resident Workers).

5.3 Fiscal Impact

Table 4: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1) log(EIT Rev)	(2) log(Total Tax)	(3) log(Total Rev)	(4) log(Police)	(5) log(Pop)	(6) log(Mkt Val)
resident_eit	0.6710**	0.2957**	0.0643	0.0531	−0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	42,309	42,526	42,543	22,979	42,523	9,323
R^2	0.977	0.983	0.959	0.953	0.997	0.978
Within R^2	0.027	0.006	0.0001	0.0001	0.0001	0.000
Fixed Effects	Municipality, Year					
Clusters	Municipality					

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Column (6) limited to 2015–2019 (market value data availability). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

5.4 Heterogeneity

6 Robustness

Table 5: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	-0.0126^{*}	-0.0104^{*}	-0.0165^{**}			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				-0.0089^{**}	-0.0079^{**}	-0.0098^{**}
				(0.0030)	(0.0028)	(0.0032)
N	27,277	14,174	27,257	3,058	2,173	3,038
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

7 Discussion

8 Conclusion

Acknowledgments

References

Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 6: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. All normalized differences < 0.15, within standard balance thresholds. Source: PA DCED Annual Financial Reports.

B Pre-Trend Tests on Fiscal Observables

C Additional Figures